



Stat3082

Introduction to Spatial Data Analysis

College of Computing

Department of Data Science

Course Outline

Second Semester, third Year

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1. Course Information

Course Code:	Stat3082	Academic Year:	Third
Course Title:	Introduction to Spatial Data Analysis	Year:	2018EC
Pre-request:		Semester:	Second
ECTS: 5 Credit Hour: 3			
Course Schedule: <i>Lecture :Wednesday,8:30 AM to 4:15 AM, Lab: Wednesday,1:30 PM to 4:15 PM</i>			
Venue: <i>Lecture: CL-03, Floor-1, Room-4 ,Lab: CL-01, Ground, Lab-14</i>			

2. Instructor Information

Course Coordinator			
Name:	Gebremedihn Desta	Email:	Gmd202@gmail.com
Telephone Number:	+251913963010	Block:	
Consultation hour:		Room:	
Course Instructor			
Name:	Haymanot Z	Email:	Haymiez@gmail.com
Telephone Number:		Block:	DH-03
Consultation hour:	Tuesday, 8:00-10:00	Room:	06

3. Course Overview/Description

This course provides a comprehensive foundation in the **concepts, techniques, and applications** of analyzing data with a spatial component. Topics include spatial data fundamentals, data visualization, exploratory spatial data analysis, spatial statistics, and interpolation methods, and practical experience with GIS and R.

4. Course Objectives

To equip students with a strong foundational understanding of spatial data analysis by developing their knowledge of spatial data concepts, visualization techniques, exploratory spatial data analysis, spatial statistics, and interpolation methods, while building practical skills using Geographic Information Systems (GIS) and R for real-world spatial problem-solving.

5. Course Learning Outcomes

At the end of the course, the student will be able to		Student Outcomes
CLO1	Understand the basic concepts of spatial data structures.	SO1,2,5
CLO2	Visualize and explore spatial data using various techniques to identify patterns and relationships in spatial data.	SO2, 5,6,7
CLO3	Utilize spatial statistics to model spatial relationships and make predictions	SO2,3,5,6,7
CLO4	Perform spatial data modeling and interpolations to estimate values at unobserved locations	SO2,3,5,6,7
CLO5	Utilize GIS and R to visualize and analyze spatial data	SO2,3,5,6,7
CLO6	Apply spatial data analysis techniques to solve real-world problems in various disciplines	SO2,3,5,6,7

6. Course Schedule and Instructional Method

This course consists of two hours of class contact hours and an additional three hours of practical hours to complete assessments, readings, and exam preparation.

Week	Topic	Learning and teaching activity	CLO
Week1	Chapter 1: Introduction 1.1 Introduction to spatial data analysis 1.11.Terminology used in spatial data analysis 1.12.Spatial data structures and formats (vector, polygon, and raster data types) 1.13.Importing and exporting spatial data 1.14.Visualization and exploration of spatial data Laboratory contents: spatial data visualization and exploration in GIS	Lecture & Practical	1 &2
Week2	Chapter 2. Spatial Data Analysis 2.1.Spatial data manipulation and preprocessing 2.2.Point pattern analysis: Clustering, spatial point processes, nearest neighbor statistics, second-order statistics, bivariate and space-time point patterns	Lecture, Practical & tutorial	1 & 2
Week3	2.3. Lattice (grid) data and Image (raster) data and analysis 2.4 Laboratory contents: all contents above using GIS		
Week4	Chapter 3. Geostatistical Analysis of Spatial Data 3.1.Introduction to geostatistics	Lecture & practical	1, 2 &3

	3.2. Spatial autocorrelation: Moran's I, Geary's C; 3.3. Isotropic and anisotropic analysis		
Week5	3.4. Variogram, semi-variogram, nugget, range, and sill 3.5. Laboratory contents: illustrate all contents using R or GIS		
Week6	Chapter 4. Spatial interpolation and prediction 4.1. Concept of spatial interpolation 4.2. Exploring different interpolation methods, such as inverse distance weighting, spline interpolation, and kriging	Lecture & practical	1,3,4 &5
Week 7	4.3. uncertainty assessment in spatial interpolation 4.4. Predictive modeling using spatial data		
Week8	4.5. Laboratory contents: illustrate different interpolation and Predictive modeling in R		
Week9	Chapter5:- Spatial autoregressive models 5.1. Concepts of spatial autoregressive models, spatial dependence, Spatial Neighbors, and spatial weights matrices	Lecture & practical	1,4 &5
Week10	5.2. Exploring different types of spatial autoregressive models, such as SAR (Spatial Autoregressive) and SEM (Spatial Error) models 5.3. Model estimation and interpretation.		
Week11	5.4. Lab: apply spatial autoregressive models in R		
Week12	Chapter 6: Applications of Spatial Statistics 6.1. Environmental science; Agriculture and forestry; Demography; Medicine and public health; Mining and petroleum engineering; Natural resources; Geography.	Lecture & practical	6
Week13	6.2. Laboratory contents: apply spatial analysis on real data applications		
Week14	Project Work or Review		

7. Assessment Tasks

No.	Assessment Method	Weight	Week Due	CLO Assessed
1.	Mid-Semester Test (written)	15	Week 6	1,2 &3
2.	Lab Practical Exams	15	Week 5	1, 2
3.	Course Project Assignment & Presentation	20	Week 12	3,4 & 5
4.	Final Exam	50	Week 15	1, 2 3, 4, 5 & 6

7.1. Submission of Assessment Tasks

Every assignment and Project must be submitted and completed by the due date. Late submissions will result in a 10%-mark loss per 24-hour delay (or portion thereof).

7.2. Feedback on Assessments

On each assignment and industrial report, feedback will be provided via email or a Telegram channel in the form of written remarks before the preceding class begins.

8. Learning Resources

8.1. Textbooks

1. Handbook_of_Spatial_Analysis_INSEE_EUROSTAT_2018

8.2. References

1. Haining, R.P., 2003. Spatial data analysis: theory and practice. Cambridge university press.
2. Bivand, R. S., Pebesma, E. J. and Gómez-Rubio, V. (2009). Applied Spatial Data Analysis with R (Use R).
3. Fischer, M.M. and Getis, A. eds., 2010. Handbook of applied spatial analysis: software tools, methods and applications (pp. 125-134). Berlin: Springer.
4. Fortin, M. J. and Dale, M. R. T. (2005). Spatial Analysis: A Guide for Ecologists.
5. Illian, J., Penttinen, A., Stoyan, H. and Stoyan, D. (2008). Statistical Analysis and Modelling of Spatial Point Patterns (Statistics in Practice).
6. Venables, W. N., and Smith, D. M. (2008). An Introduction to R: Notes on R – A Programming Environment for Data Analysis and Graphics, Version 2.6.2.

9. Grading system

Examinations are graded on letter grading system as stated in the active Senate legislation of DBU. The status description is based on the raw mark interval given in Table 3.

Raw Mark interval (100%)	Corresponding Letter Grade	Corresponding fixed number Grade	Status Description	Class Description
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[90,100]	A+	4.0	Excellent	First Class with Great Distinction
[85,90)	A	4.0		
[80,85)	A-	3.75		
[75,80)	B+	3.5	Very Good	First Class with Distinction
[70,75)	B	3.0		
[65,70)	B-	2.75	Good	First Class
[60,65)	C+	2.5		Second Class
[50,60)	C	2.0	Satisfactory	
[45,50)	C-	1.75	Unsatisfactory	Lower Class
[40,45)	D	1.0	Very Poor	
[0,40)	F	0	Fail	Lowest Class

10. Pass Requirements

To pass this course a student must:

- Students must make a genuine attempt at every piece of assessable work, including the mid-semester test, lab practical exams, course project, and the final examination. Failure to submit or attend any assessment (without an officially approved, documented reason such as a medical emergency) will result in a failing grade for the course.
- A student must achieve a score of **at least 40% on the final examination**. This condition ensures that passing students have demonstrated a foundational, individual understanding of the core course material, independent of other graded work. Note: Even if a student's overall composite mark is above 50%, failure to meet this minimum exam requirement will result in a failing grade (F).

11. Attendance

As per university guidelines, a minimum of 80% attendance during lectures and 100% during practical work sessions is required, except for some unprecedented mishaps. Failure to fulfil this requirement will result in barring from examinations.

12. Academic Integrity, Referencing and Plagiarism

12.1. Academic Integrity

DBU values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism, and other academic offenses under the Code of Student Conduct and Disciplinary Procedures. Work submitted must be your own, and others' ideas should be appropriately acknowledged. If you don't follow these rules, plagiarism may be detected in your work using the plagiarism detection tool 'Turnitin'.

12.2. Referencing

Referencing is a way of acknowledging the sources of information that you use to research your assignments. You need to provide a reference whenever you draw on someone else's words, ideas, or research. Not referencing other people's work can constitute plagiarism.

12.3. Referencing Style

The University advises students to use the IEEE referencing style for written work and oral presentations.